

Web Development 3.2 Final Project

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Programme: Diploma in Information Technology

Subject: Web Development 3.2

Project Title: Functional and Compliant Portfolio Website

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1 Project Overview

This project demonstrates the development of a **fully functional, responsive, and compliant portfolio website**.

It showcases web design and development best practices using **HTML5, CSS3, and JavaScript* to meet industry standards.

The website consists of:

- **Home Page** – Introduces the portfolio and includes a navigation, a hero section, a small portfolio section, and text logo on the nav bar.
 - **About Page** – Displays my profile, technical skills and competencies, and my career goals.
 - **Contact Page** – Includes a working contact form using HTML.
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2 Objectives

- Apply modern web technologies (HTML, CSS, JS).
 - Demonstrate responsive design principles for mobile and desktop.
 - Implement clean and accessible UI elements.
 - Use JavaScript for interactivity and theme control.
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3 Technologies Used

Technology	Purpose
HTML5	Structure and accessibility.
CSS3	Styling, responsive layout, and animations.
JavaScripts	Theme switch, scroll reveal, and mobile navigation.
SVG	Lightweight dynamic logo animation.
LocalStorage	Saves user theme preference.

4 Design & Structure

- **Color Scheme:** Light/dark mode using CSS variables.
 - **Typography:** "Poppins" system font for modern readability.
 - **Layout:** Flexbox and media queries ensure responsive layouts.
 - **Consistency:** Reused header/footer across all pages with working navigation and hover states.
 - **Interactivity:**
 - Theme toggle (manual switch).
 - Smooth page fade and scroll reveal effects using css + js.
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5 Functionality

Feature	Description
Navigation	Sticky header, hover effects, and mobile hamburger menu.
Responsiveness	Mobile-first design using media queries (@media on css).
Animations	Page load fade-in + scroll-based reveal.
Theme	Allows manual switching.

6 Challenges & Solutions

Challenge	Solution
JavaScript not detecting menu elements	Added correct element IDs (menu-toggle, nav-list) and deferred script loading.
Responsive layout adjustments	Applied media queries and Flexbox-based containers for mobile users.
Styling not applied	Had to fix css variables and add additional variables

9 Conclusion

This portfolio website represents the practical application of web design and development principles learned in **Web Dev 3.1 & 3.2**.

It is functional, visually engaging, responsive, and demonstrates industry-standard coding practices ready for real-world application.

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